

light bounces off it.

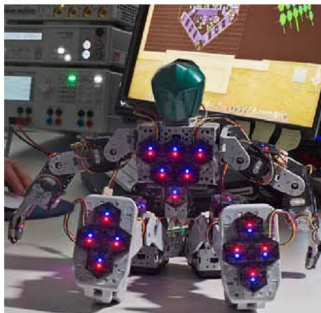
The researchers engineered a structure in which they etched tiny holes, creating a material known as a 'photonic crystal,' which behaves as though it has zero index - light can travel with an ultra-fast velocity in this environment. The material, a coating no thicker than one-hundredth of the diameter of a strand of hair, has properties that don't occur in nature.

When light travels, it bends—in technical terms, it disperses and incurs 'phase,' an oscillating curve that leaves a trail of information behind it. Those oscillations show an object's properties, such as shape and size, which can identify it. However, light hits this specially engineered material without leaving a trace.

"We can now control the flow of light, the fastest thing known to us. This can enable self-focusing light beams, highly directive antennas, and even potentially an approach to hide objects, at least in the small scale or a narrow band of frequencies," said Wong.

Sensitive skin for robots

Scientists at the Institute of Cognitive Systems at Technical University of Munich (Technische Universität München, TUM) are developing an artificial skin for robots. The skin



Bioid robot with 31 hexagonal sensor modules

will provide important tactile information to the robot and thus supplement its perception formed by camera eyes, infrared scanners and gripping hands. The artificial skin when touched could, for example, lead to a spontaneous retreat (when the robot hits an object) or cause the machine to use its eyes for the first time to search for the source of contact.

The centre-piece of the robotic shell is a 5cm² hexagonal plate or circuit board. Each small circuit board contains four infrared sensors that detect anything closer than 1 cm. "We

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- Provides sharper visual clarity than fluorescent tube
- Life of 12 to 15 years, with a warranty of 5 years



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